Supplementary D — Extended results

Companion to: *A Band-Mask Robustness Diagnostic for LDA on HSI*

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A.1 Scope

This supplement reports the full per-(scene, mask) outputs of the diagnostic that did not fit in the 4-page conference paper. All numbers below are read directly from the public artefact `data/derived/band_masks/ca` for the labelled-scene side, and from `data/derived/band_masks_hidsag/index.json` plus the per-(subset, mask) `summary.json` for HIDSAG.

B.2 Full labelled-scene table

Table 1 reports all five reportable quantities per (scene, mask) tuple, in addition to the paired ARI already shown in Table I of the main paper.

C.3 Hungarian alignments (canonical → masked)

Table 2 reports the full Hungarian topic-id permutation σ^* per (scene, mask). Each row gives the mapping $\sigma^*(0), \sigma^*(1), \ldots, \sigma^*(K-1)$, i.e. for each canonical topic id the masked-topic id it is matched to.

D.4 HIDSAG companion results

HIDSAG has no per-pixel ground truth, so the paired ARI metric does not apply directly. The diagnostic on HIDSAG reports per-(subset, mask) the train perplexity, the mean per-document max-θ (a confidence proxy), and the bands kept; Table 3 reports the full 5×4 grid. The two extreme subsets are MINERAL2 (very high confidence, 0.85–1.00 across masks, signalling sharp topic identities) and GEOCHEM (low confidence, 0.29–0.39, signalling diffuse topic identities).

E.5 Reading the swap-rate column

The swap rate under Hungarian alignment is, by definition, ≥ 0 and $\leq 1 - 1/K$ (a perfectly random permutation under K balanced clusters reaches a swap rate of $1 - 1/K$). On the labelled $K = 12$ scenes the upper bound is ~ 0.917. The values observed:

- Indian Pines: swap rates 0.45–0.53 across masks; the canonical basis is partially robust.
- Salinas: swap rates 0.35–0.44; similar to Indian Pines.
- Salinas-A: swap rate as low as 0.18 under SWIR; this is the strongest single recovery in the entire diagnostic.
- KSC: swap rates 0.53–0.64; band-fragile across masks.

Table 1: Extended per-(scene, mask) outputs. Columns: paired ARI of canonical vs masked dominant-topic maps (P-ARI); swap rate under Hungarian alignment of canonical σ^* (Swap); number of bands kept after the mask, with sensor floor B in parentheses; train perplexity of the masked LDA refit (Perp); ARI of the masked dominant-topic map against the ground-truth label (ARI-lbl). Pavia U $\times$ SWIR is auto-skipped (sensor is VNIR-only, $V^{(m)} = 0$).

Scene	Mask	P-ARI	Swap	Bands kept	Perp	ARI-lbl
Indian Pines	VNIR	0.427	0.525	67/200	64.34	0.131
	SWIR	0.527	0.446	133/200	131.21	0.165
	no-water	0.499	0.484	177/200	167.45	0.198
	top-50 F.	0.398	0.505	50/200	54.28	0.144
Salinas	VNIR	0.540	0.348	68/204	64.25	0.136
	SWIR	0.435	0.443	136/204	127.73	0.094
	no-water	0.502	0.345	180/204	162.94	0.135
	top-50 F.	0.507	0.371	50/204	48.03	0.125
Salinas-A	VNIR	0.521	0.357	68/204	66.78	0.293
	SWIR	**0.766**	**0.176**	136/204	95.22	0.432
	no-water	0.533	0.368	180/204	148.26	0.268
	top-50 F.	0.352	0.388	50/204	51.39	0.174
Pavia U	VNIR	0.415	0.414	103/103	100.57	0.316
	SWIR	—	—	0/103	—	—
	no-water	0.415	0.414	103/103	100.57	0.316
	top-50 F.	0.387	0.471	50/103	51.38	0.318
KSC	VNIR	0.009	0.636	59/176	56.61	0.220
	SWIR	0.011	0.547	117/176	103.21	0.124
	no-water	0.007	0.637	155/176	136.42	0.270
	top-50 F.	0.013	0.527	50/176	53.68	0.098
Botswana	VNIR	0.108	0.654	49/145	50.98	0.125
	SWIR	0.013	0.749	97/145	100.62	0.098
	no-water	0.127	0.640	127/145	127.06	0.200
	top-50 F.	0.117	0.636	50/145	54.16	0.127

- Botswana: swap rates 0.64–0.75; the most extreme band-fragile scene in the diagnostic.

Note that the swap rate and the paired ARI agree on the band-fragile/band-robust split: scenes with low paired ARI also have high swap rate. The two metrics are complementary, however: ARI corrects for chance given the marginal sizes, swap rate is a direct fraction of mis-assigned pixels, and the two diverge when one fit's marginal is strongly imbalanced (e.g. Salinas-A SWIR's prevalence $[0.188, 0.124, 0.094, 0.365, 0.187, 0.043]$ is closer to balanced than KSC's, so the chance correction is smaller, so the ARI value is closer to the swap-rate complement).

References

Table 2: Hungarian permutation σ^* per (scene, mask). Row (s, m) lists the matched masked-topic id for each canonical-topic id in turn (entries separated by commas). Identity rows (canonical topic k matches masked topic k) indicate a stable column.

Scene	Mask	$\sigma^*(0), \sigma^*(1), \ldots$
Indian Pines	VNIR	8, 10, 1, 3, 4, 9, 11, 7, 6, 2, 0, 5
Indian Pines	SWIR	6, 7, 2, 0, 4, 11, 3, 5, 9, 1, 8, 10
Indian Pines	no-water	3, 1, 8, 0, 2, 9, 6, 11, 5, 4, 10, 7
Indian Pines	top-50 F.	0, 7, 11, 2, 1, 8, 4, 3, 5, 6, 9, 10
Salinas	VNIR	1, 5, 6, 7, 2, 0, 9, 8, 3, 4, 10, 11
Salinas	SWIR	5, 11, 1, 2, 7, 0, 3, 9, 4, 8, 6, 10
Salinas	no-water	3, 1, 8, 6, 5, 0, 2, 10, 4, 9, 7, 11
Salinas	top-50 F.	7, 3, 1, 6, 2, 0, 10, 4, 9, 5, 8, 11
Salinas-A	VNIR	4, 1, 5, 3, 2, 0
Salinas-A	SWIR	2, 5, 0, 4, 1, 3
Salinas-A	no-water	3, 4, 5, 0, 1, 2
Salinas-A	top-50 F.	2, 0, 1, 5, 3, 4
Pavia U	VNIR	0, 1, 6, 3, 5, 7, 8, 4, 2
Pavia U	SWIR	— (skipped)
Pavia U	no-water	0, 1, 6, 3, 5, 7, 8, 4, 2
Pavia U	top-50 F.	0, 1, 2, 3, 4, 5, 6, 8, 7
KSC	VNIR	7, 5, 1, 6, 0, 2, 3, 4, 11, 8, 9, 10
KSC	SWIR	3, 2, 7, 6, 0, 1, 4, 5, 11, 8, 9, 10
KSC	no-water	7, 0, 2, 8, 1, 3, 10, 5, 4, 9, 6, 11
KSC	top-50 F.	4, 11, 2, 10, 6, 1, 8, 5, 0, 7, 3, 9
Botswana	VNIR	0, 10, 1, 11, 9, 3, 7, 8, 6, 4, 2, 5
Botswana	SWIR	1, 2, 5, 8, 6, 3, 4, 0, 11, 9, 10, 7
Botswana	no-water	11, 3, 5, 7, 8, 1, 4, 0, 2, 9, 10, 6
Botswana	top-50 F.	1, 4, 2, 3, 9, 0, 6, 7, 8, 11, 10, 5

Table 3: HIDSAG band-mask sweep — perplexity, mean confidence (per-doc max-θ), and bands kept. Sensor floor $B = 268$ on the `swir_low` modality.

Subset	Mask	Perp	Conf	Bands
GEOMET	VNIR	20.46	0.354	18
	SWIR	259.62	0.298	250
	no-water	228.77	0.384	222
	top-50 var.	276.92	0.355	268
MINERAL1	VNIR	20.02	0.365	18
	SWIR	257.92	0.416	250
	no-water	229.03	0.409	222
	top-50 var.	53.14	0.298	50
MINERAL2	VNIR	19.52	**0.946**	18
	SWIR	260.68	0.845	250
	no-water	228.27	**0.997**	222
	top-50 var.	278.34	0.994	268
GEOCHEM	VNIR	19.95	0.289	18
	SWIR	258.57	0.374	250
	no-water	228.72	0.391	222
	top-50 var.	53.20	0.385	50
PORPHYRY	VNIR	19.72	0.487	18
	SWIR	257.90	0.397	250
	no-water	229.19	0.405	222
	top-50 var.	52.93	0.437	50